Universidade do Minho

Ano Letivo: 2025/26

Bases de Dados NoSQL

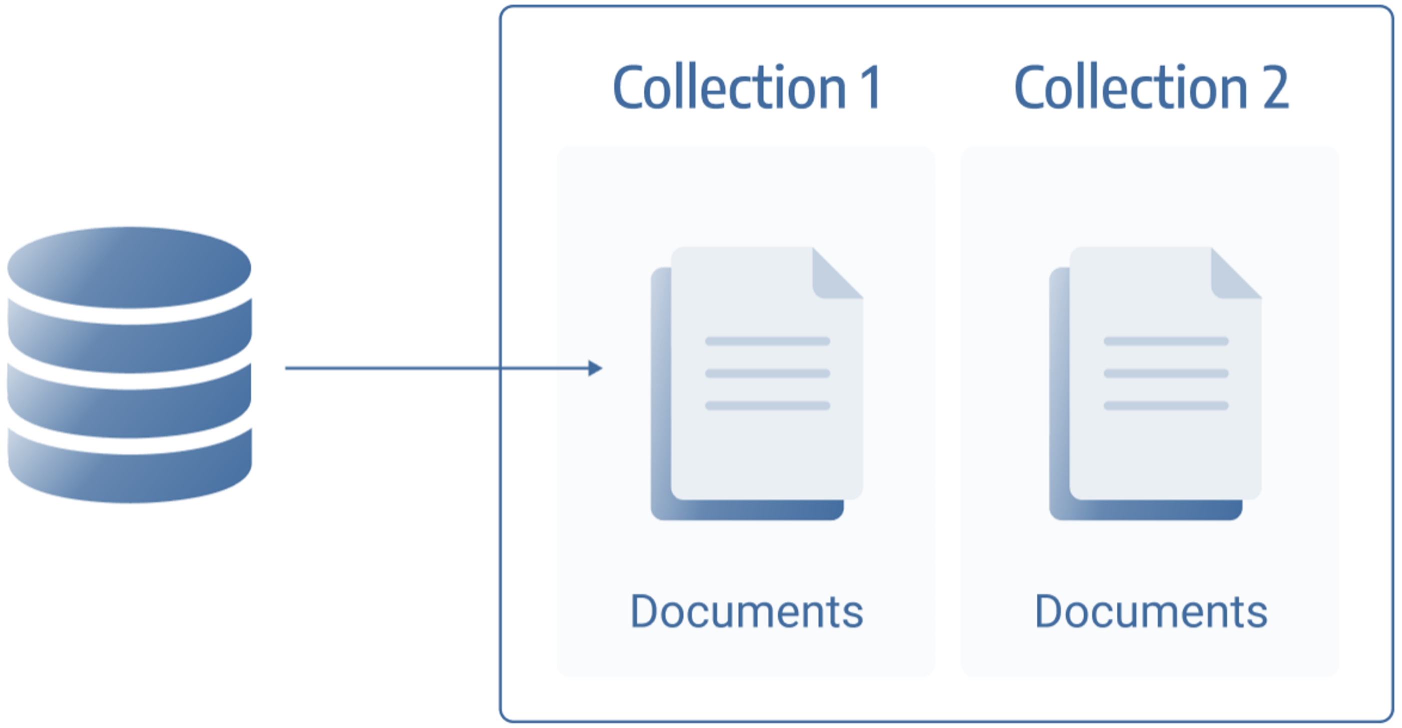
T03 – Bases de Dados Documentais

02

Document Store Database

What is it?

- NoSQL database;
- Data is stored as documents (JSON, BSON, XML, ...);
- Flexible and scalable structure.



Document Store Database

What is it?

- Document databases store each record and all associated data in a single document;
- Each document consists of semi-structured data;
- This data can be queried using various DBMS query and analysis tools.

Document Store Database

Examples of documents

Next there are two examples of documents that can be stored in a document database.

The same data is presented in different languages.

Document Store Database

Examples of documents

XML.

```
<artist>
  <artistname>Iron Maiden</<artistname>
  <albums>
    <album>
      <albumname>The Book of Souls</albumname>
      <datereleased>2015</datereleased>
      <genre>Hard Rock</genre>
    </album>
    <album>
      <albumname>Killers</albumname>
      <datereleased>1981</datereleased>
      <genre>Hard Rock</genre>
    </album>
    <album>
      <albumname>Powerslave</albumname>
      <datereleased>1984</datereleased>
      <genre>Hard Rock</genre>
    </album>
    <album>
      <albumname>Somewhere in Time</albumname>
      <datereleased>1986</datereleased>
      <genre>Hard Rock</genre>
    </album>
  </albums>
</artist>
```

02

Document Store Database

Examples of documents

JSON.

```
{
	'_id' : 1,
	'artistName' : { 'Iron Maiden' },
	'albums' : [
		{
			'albumname' : 'The Book of Souls',
			'datereleased' : 2015,
			'genre' : 'Hard Rock'
		}, {
			'albumname' : 'Killers',
			'datereleased' : 1981,
			'genre' : 'Hard Rock',
		}, {
			'albumname' : 'Powerslave',
			'datereleased' : 1984,
			'genre' : 'Hard Rock'
		}, {
			'albumname' : 'Somewhere in Time',
			'datereleased' : 1986,
			'genre' : 'Hard Rock'
		}
	]
}
```

Document Store Database

Examples of documents

- We added the _id field in the second example (JSON).
- This is not always mandatory.
- Some DBMSs automatically insert a unique ID.

Document Store Database

Examples of documents

- We added the _id field in the second example (JSON).
- This is not always mandatory.
- Some DBMSs automatically insert a unique ID.

Document DB vs RDBMS

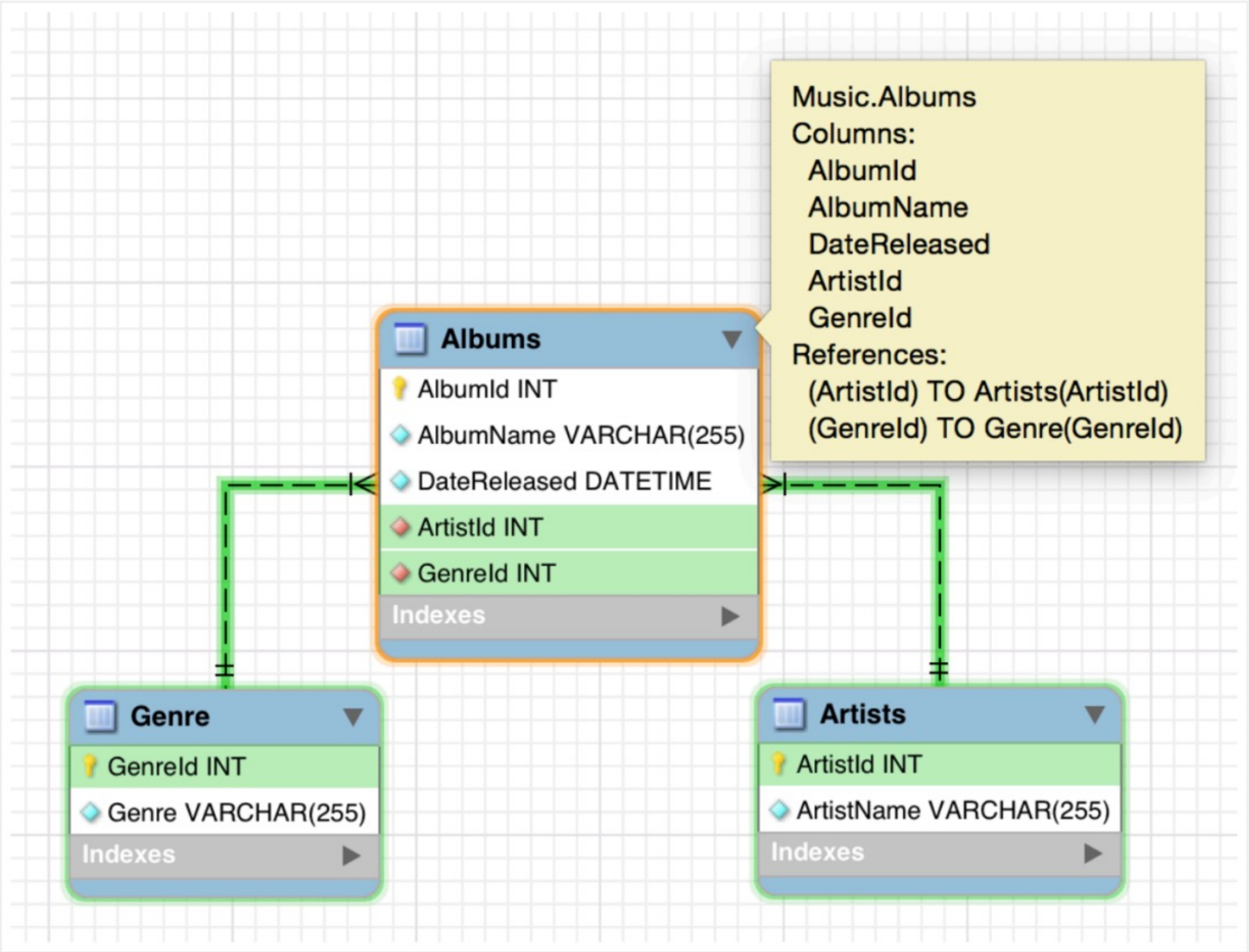
Example

If we insert the presented data into a relational database, normalizing the information will require using three different tables.

These tables will be related through primary key and foreign key attributes.

Document DB vs RDBMS

➡ Example



Document DB vs RDBMS

Example

Artists

ArtistId	ArtistName
1	Iron Maiden
2	Devin Townsend
3	The Wiggles
4	...

Albums

AlbumId	AlbumName	DateReleased	ArtistId	GenreId
1	The Book of Souls	2015	1	3
2	Killers	1981	1	3
3	Powerslave	1984	1	3
4	Somewhere in Time	1986	1	3
5	Ziltoid the Omniscient	2007	2	3
6	...	...	...	...

Genre

GenreId	Genre
1	Country
2	Blues
3	Hard Rock
4	...

Document DB vs RDBMS

Terminology

RDBMS	Document DB
Table	Collection
Row	Document
Primary Key	_id
Foreign Key	Embedded Document

Document DB vs RDBMS

Tables

Relational databases store data in tables.
Each table contains columns, and each row represents a record.
Information about an entity can be spread across multiple tables.
Data from different tables can be associated by establishing relationships between them.

Document databases, on the other hand, do not use tables.
Data for a particular entity is stored in a single document, and all associated data is contained within that document.

Document DB vs RDBMS

Schemas

In relational databases, we create a logical schema before inserting any data.

In document databases (a common characteristic of most NoSQL databases), this requirement does not exist.
We can simply load all the data without using any predefined schema.

Document DB vs RDBMS

➔ Scalability

Document databases can be scaled horizontally.

Data can be stored across thousands of computers, and the system will still maintain good performance. This process is known as **sharding**.

Sharding involves distributing data across multiple machines to improve scalability and performance.

Document DB vs RDBMS

Scalability

Relational databases are not suitable for this type of use.

Relational databases are better suited for vertical scaling (i.e., adding more memory, disk space, etc.).

However, there is always a limit to the resources of a single machine. In such cases, horizontal scalability may be the only viable option.

Document DB vs RDBMS

➡ Relationships

Data stored in document databases do not have foreign keys.

Foreign keys are used in relational databases to enforce relationships between tables.

In document databases, relationships are typically handled at the application level, rather than being enforced by the database itself.

Document DB vs RDBMS

Relationships

The idea behind the document model is that all data associated with a record is stored in the same document.

The need to establish a relationship in the document model does not have the same impact as in a relational database.

Document DB vs RDBMS

NoSQL

Relational databases use SQL as the query language.

Document databases tend to use other query languages (although some provide support for SQL).

Most offer languages like XQuery, XSLT, SPARQL, Java, JavaScript, Python, etc.

Document DB vs RDBMS

NoSQL

Relational databases use SQL as the query language.

Document databases tend to use other query languages (although some provide support for SQL).

Most offer languages like XQuery, XSLT, SPARQL, Java, JavaScript, Python, etc.

Document Model Stores

Usage Examples

Web Applications

- Content management systems
- Blogging platforms
- eCommerce applications
- Web analytics
- User preferences data

Document Model Stores

Usage Examples

User Generated Content

- Chat sessions
- Tweets
- Blog posts
- Ratings
- Comments

Document Model Stores

Usage Examples

Catalog Data

- User accounts
- Product catalogs
- Device registries for Internet of Things
- Bill of materials systems

Document Model Stores

Usage Examples

Gaming

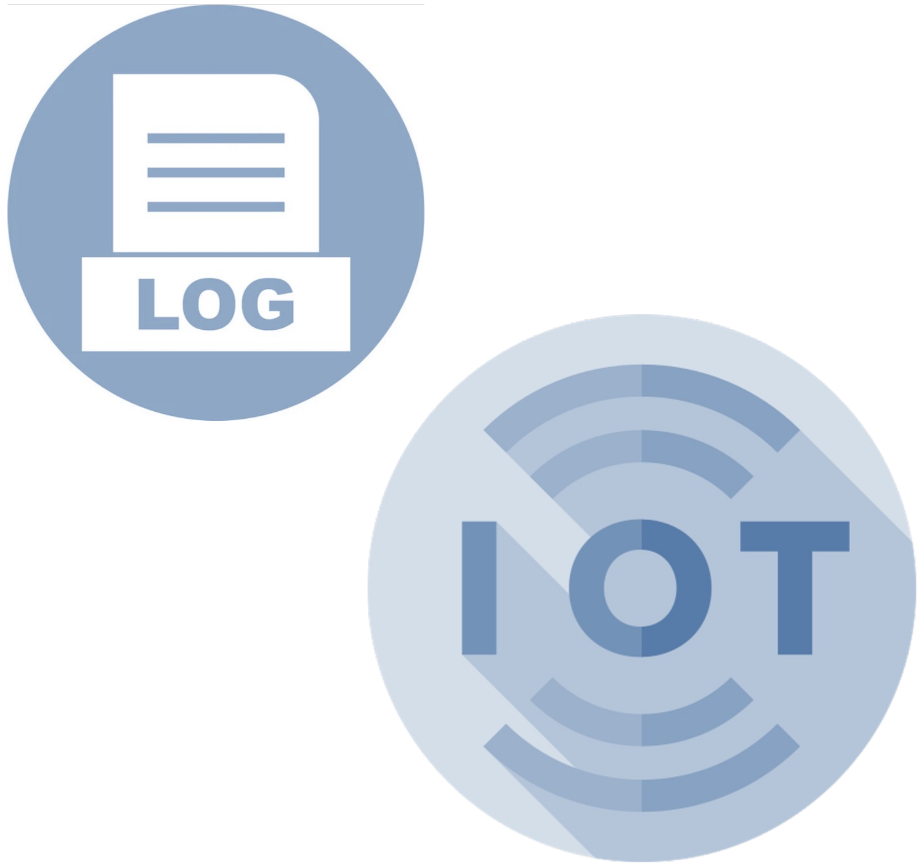
- In-game stats
- Social media integration
- High-score leaderboards
- In-game chat messages
- Player guild memberships
- Challenges completed

Document Model Stores

Usage Examples

Networking/computing

- Sensor data from mobile devices
- Log files
- Realtime analytics
- Various other data from Internet of Things



Document Model Stores

Examples

mongoDB®

Amazon DocumentDB

CouchDB

MarkLogic™

OrientDB®

TOP 12 de Document Model Stores

https://www.predictiveanalyticstoday.com/top-nosql-document-databases/

Universidade do Minho

Ano Letivo: 2025/26

Bases de Dados NoSQL

T03 – Bases de Dados Documentais